EN3160 – Assignment 02

Name: Saruka U. Index No: 230585C

GitHub: https://github.com/Saruka03/EN3160-Assignment-02

1. Blob Detection in the Sunflower Image

The sunflower image was processed using a scale-normalized Laplacian of Gaussian (LoG) detector. The scale-space was evaluated over 20 scales in the range $\sigma = 2.2$ to $\sigma = 12.0$. Local maxima were selected using a response threshold of 12% of the maximum response. Detections with $\sigma < 2.9$ were excluded, and spatial suppression with a 10-pixel centre distance was applied.

The final experiment retained 185 blob detections. These include sunflower heads as well as some background structures; therefore, the number of detections is not an exact sunflower count.

The five largest detected circles, ranked by radius, are listed in Table . The approximate radius was calculated using $r = \sqrt{2}\sigma$.

Centre (x, y)	Radius (pixels)	σ	Response
$(311, 198)$	16.97056	12.00000	0.10385
$(123, 359)$	16.97056	12.00000	0.12126
$(195, 295)$	16.97056	12.00000	0.12687
$(261, 208)$	16.97056	12.00000	0.10860
$(135, 276)$	16.97056	12.00000	0.08139



Figure 1: Detected blobs in the sunflower image.

The detector retained 185 blobs over 20 scales in the range $\sigma = 2.2$ to $\sigma = 12.0$. The largest detected radius was 16.97 pixels at $\sigma = 12.0$. Since the largest detections occur at the maximum tested scale, some circles may represent larger image structures rather than individual sunflowers.

2. RANSAC Line and Circle Fitting

RANSAC was implemented to estimate a line first, followed by a circle fitted to the points remaining after removing the line inliers. The line inlier threshold was 1.4, and the circle radial-error threshold was 1.2.

The estimated line parameters were

$$a = 0.716670, \quad b = 0.697413, \quad d = 1.502897.$$

Therefore, the fitted line is

$$0.716670x + 0.697413y = 1.502897.$$

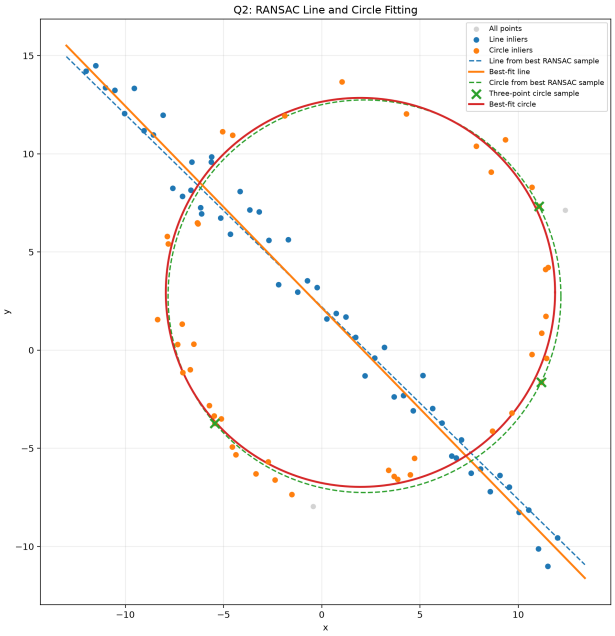
The unit-normal constraint was satisfied: $\sqrt{a^2 + b^2} = 1.000000$. The final line contained 54 inliers out of 100 points.

The estimated circle centre and radius were

$$(c_x, c_y) = (1.969390,\ 2.942512), \qquad r = 9.907175.$$

The circle contained 44 inliers. These estimates are close to the ground-truth centre $(2, 3)$ and radius 10.

The plot shows the best sample models, the refined models, and the detected inlier sets. Some points may be assigned to the wrong model because of noise and spatial overlap.



What if the circle were fitted first?

RANSAC could first sample three points to estimate candidate circles and select the model with the largest radial-error consensus. After removing the circle inliers, line fitting could be performed on the remaining points. This reversed order can also work, although the result depends on sampling, noise, and the chosen thresholds.

Important implementation details

The line RANSAC loop samples two points and counts inliers using perpendicular distance. The circle RANSAC loop samples three points and counts inliers using radial error. The best models are refined using total least squares for the line and squared radial-error minimization for the circle.

```
# Line inlier test
line_error = np.abs(X @ np.array([a, b]) - d)
line_inliers = line_error < 1.4

# Circle inlier test
radial_error = np.abs(
    np.linalg.norm(points - center, axis=1) - radius
)
circle_inliers = radial_error < 1.2
```